

OLIST E-COMMERCE

Power BI Technical Documentation

End-to-End Customer Journey Intelligence

Project: Project depi
Dataset: Brazilian E-Commerce (Olist) · 2016–2018
Pages: 4 Dashboard Pages + Data Model + Full DAX Library
Total Measures: 28 DAX Measures · 7 Calculated Columns
Records: 99,441 Orders · 3,095 Sellers · 99,441 Customers

Project Team

01 Rawda Hesham	02 Shaza Yasser
03 Ismail Mahmoud	04 Hamza Ahmed
05 Omar Zakria	06 Wadie Tofiek

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1. Project Overview & Business Context

This Power BI report delivers a full end-to-end view of Olist's Brazilian e-commerce operations — covering revenue performance, logistics efficiency, product category profitability, and customer behavior. The project transforms 8 raw CSV files into an interactive 4-page dashboard that supports strategic and operational decision-making.

1.1 Business Objectives

- Identify growth drivers, top regions, and top product categories
- Detect delivery delays, seller performance issues, and logistics inefficiencies
- Understand customer satisfaction drivers and loyalty patterns
- Enable data-driven decisions for seller management, logistics, and marketing

1.2 Report Structure

Page	Title	Primary Focus	Key KPIs
1	Commercial Overview	Revenue & Market Performance	Revenue, AOV, On-Time %, Total Orders, Avg Review
2	Operations Intelligence	Delivery & Logistics	Active Sellers, On-Time %, Avg Delivery, Delayed Orders, Top Route
3	Category Analysis	Product & Freight Performance	Freight Ratio, Avg Freight, Avg Price, Total Orders, High Freight %
4	Customer Insights	Loyalty & Demographics	Total Customers, Avg CLV, Orders/Customer, Avg Review

2. Data Sources & Raw Files

The dataset is the publicly available Olist Brazilian E-Commerce dataset. Eight CSV files were loaded into Power BI.

File Name	Rows	Role in Model	Key Columns
olist_orders_dataset.csv	99,441	Fact (bridging)	order_id, customer_id, order_status, timestamps
olist_order_items_dataset.csv	112,650	Fact (main — fact_order_items)	order_id, product_id, seller_id, price, freight_value
olist_order_payments_dataset.csv	103,886	Fact (fact_payment)	order_id, payment_type, payment_value, installments
olist_order_reviews_dataset.csv	98,673	Fact (fact_reviews)	order_id, review_score, timestamps
olist_customers_dataset.csv	99,441	Dimension (dim_customer)	customer_id, customer_unique_id, city, state

olist_sellers_dataset.csv	3,095	Dimension (dim_sellers)	seller_id, seller_name, city, state
olist_products_dataset.csv	32,951	Dimension (dim_product)	product_id, category, weight_g, dimensions
product_category_name_translation.csv	71	Lookup (merged into dim_product)	category PT → EN translation

3. Data Model & Relationships (Star Schema)

The model follows a Star Schema design with fact_order_items as the central fact table, surrounded by dimension tables. A dedicated Date Dim table is created via DAX to enable time intelligence functions.

3.1 Tables in the Model

Table Name	Type	Description
fact_order_items	Fact	Central fact table — one row per order item. Contains price, freight_value, seller_id, product_id, timestamps, and all calculated columns.
fact_payment	Fact	Payment details per order — payment_type, payment_value, installments.
fact_reviews	Fact	Customer reviews — review_score per order_id.
dim_order	Dimension	Order-level metadata — order_status, customer_id.
dim_customer	Dimension	Customer profile — customer_unique_id, city, state, zip_code.
dim_sellers	Dimension	Seller profile — seller_id, seller_name, city, state.
dim_product	Dimension	Product specs — category (English), weight_g, dimensions.
dim_geolocation	Dimension	Geographic data — lat, lng, city, state, zip_code.
Date Dim	Date Table	Created via DAX (CALENDARAUTO). Contains Date, Day Number, Day of Week, Is Weekend, Month Name.
Dax Measures	Measure Table	Empty table used to host all DAX measures — best practice for measure organization.
Parameter	Parameter Table	What-if parameter — used for dynamic Top N filtering on visuals.

3.2 Relationships

From Table · Column	To Table · Column	Cardinality	Active
fact_order_items [order_id]	dim_order [order_id]	Many → One	✓ Active
fact_order_items [product_id]	dim_product [product_id]	Many → One	✓ Active

fact_order_items [seller_id]	dim_sellers [seller_id]	Many → One	✓ Active
dim_order [customer_id]	dim_customer [customer_id]	Many → One	✓ Active
dim_customer [zip_code]	dim_geolocation [zip_code]	Many → One	✓ Active
fact_payment [order_id]	dim_order [order_id]	Many → One	✓ Active
fact_reviews [order_id]	dim_order [order_id]	Many → One	✓ Active
Date Dim [Date]	fact_order_items [order_date]	One → Many	✓ Active (default)
Date Dim [Date]	fact_order_items [order_delivered_timestamp]	One → Many	⚠ Inactive*

* Inactive relationship activated inside On-Time Delivery % measure via USERELATIONSHIP(Date Dim[Date], fact_order_items[order_delivered_timestamp])

⚠ Important: The Date Dim table is marked as a Date Table in Power BI (right-click → Mark as date table). This is required for all time intelligence DAX functions (DATESYTD, SAMEPERIODLASTYEAR, DATEADD, etc.) to work correctly.

⚠ The relationship between Date Dim and fact_order_items[order_delivered_timestamp] is INACTIVE by default.

It is activated only inside the On-Time Delivery % measure using USERELATIONSHIP().

3.3 Date Dim — DAX Definition

```
Date Dim =
    ADDCOLUMNS(
        CALENDARAUTO(),
        "Day Number",    DAY([Date]),
        "Day of Week",   FORMAT([Date], "dddd"),
        "Is Weekend",    IF(WEEKDAY([Date], 2) >= 6, TRUE(), FALSE()),
        "Month Name",    FORMAT([Date], "MMM YYYY")
    )
```

4. Data Transformations & Power Query

4.1 Transformations Applied

Table	Transformation	Reason
fact_order_items	Merged olist_orders columns (order_status, customer_id, timestamps) into fact_order_items	Centralize all order-level data for calculated columns
fact_order_items	Renamed order_purchase_timestamp → order_date	Cleaner field name for Date Dim relationship

fact_order_items	Renamed order_delivered_customer_date → order_delivered_timestamp	Consistency in naming
fact_order_items	Renamed order_estimated_delivery_date → deadline_date	Clarity for delivery calculations
dim_product	Merged product_category_name_translation into products table	English category names for all visuals
dim_product	Renamed product_category_name_english → category	Shorter field name
dim_sellers	Added seller_name column (synthetic — sellers identified by state + id)	Required for seller table visual on Page 2
dim_customer	Promoted headers, set data types (text/number)	Data integrity
All tables	Removed null rows in key columns (order_id, product_id)	Data quality
All tables	Set correct data types for timestamp columns → Date/Time	Required for DATEDIFF calculated columns

5. Calculated Columns

All calculated columns live in the fact_order_items table unless noted. They are computed row-by-row at data refresh time and stored in the model.

5.1 Delay Days

Calculates the delay in days between actual delivery date and the promised estimated delivery date. Positive = late, Negative = early, Zero = exact.

```
// Table: fact_order_items
Delay Days =
    IF(
        fact_order_items[order_status] = "delivered",
        DATEDIFF(
            fact_order_items[order_estimated_delivery_date],
            fact_order_items[order_delivered_timestamp],
            DAY
        )
    )
// Result: positive = late, negative = early, BLANK for non-delivered orders
// Avg when late: +10.1 days | Max: 189 days
```

5.2 Delay Severity

Categorizes each order's delay into 4 severity buckets for use in charts and slicers.

```
// Table: fact_order_items
Delay Severity =
    SWITCH(
        TRUE(),
        [Delay Days] <= 0, "On Time",
        [Delay Days] <= 3, "Low Delay",
```

```

        [Delay Days] <= 7, "Medium Delay",
        "Critical Delay"
    )
// Used in: bar chart on Page 2 – Avg Delay Duration by state

```

5.3 Delivery Status

```

// Table: fact_order_items
Delivery Status =
    IF(
        fact_order_items[Delay Days] > 0,
        "Delayed",
        "On Time"
    )
// Binary flag for filtering – used in KPI card for Delayed Orders

```

5.4 Late Delivery Flag

```

// Table: fact_order_items
Late Delivery Flag =
    IF(
        fact_order_items[Delay Days] > 3,
        1,
        0
    )
// Numeric flag (1/0) – threshold >3 days for 'meaningfully late'
// Used in: Delayed Orders measure, On-Time % measure

```

5.5 Route

Creates a seller-to-customer route string by combining seller state and customer state. Used to identify the most delay-prone shipping routes.

```

// Table: fact_order_items
Route =
    RELATED(dim_sellers[state])
    & " -> " &
    RELATED(dim_customer[state])
// Example: "SP -> BA" | "SP -> RJ" | "BA -> MA"
// Used in: Top Delay Route measure (Page 2 KPI card)

```

5.6 Freight Ratio %

```

// Table: fact_order_items
Freight Ratio % =
    DIVIDE(
        fact_order_items[freight_value],
        fact_order_items[Total_cost]
    )
// Row-level freight ratio: freight ÷ (price + freight)
// Note: Total_cost must exist as a column: price + freight_value

```

5.7 delivery_days (lowercase)

```
// Table: fact_order_items (separate from Delay Days above)
delivery_days =
    DATEDIFF(
        fact_order_items[order_date],
        fact_order_items[order_delivered_timestamp],
        DAY
    )
// This measures TOTAL journey time (purchase → delivery)
// Different from Delay Days which measures vs estimated date
// Used in: Avg Delivery Time KPI (Page 2) = 12.43 days
```

6. DAX Measures — Complete Library

All 28 measures are stored in the dedicated 'Dax Measures' table (empty table — best practice). Below is each measure with its DAX code, output value, and usage.

Measure Name	Verified Output	Page Used	Category
(AOV)	160.99 BRL	1, all pages	Revenue
Active Sellers	3,095 / 2,970 active	2	Operations
Average CLV	141.44 BRL	4	Customer
Avg Delay by Route	Varies by route	2	Operations
Avg Delay Duration	10.1 days (late only)	2	Operations
Avg Delivery Time	12.43 days	2	Operations
AVG Freight Cost	19.99 BRL	3	Category
Avg Orders Per Customer	1.03	4	Customer
Avg Product Price	145.30 BRL	3	Category
Avg Review Score	4.09 / 5	1, 2, 4	Customer
Black Friday Peak Marker	Revenue on Nov 24 2017	1	Revenue
Delayed Orders	7,000+ orders	2	Operations
Freight Cost	2.25M BRL total	3	Category
Freight Cost Ratio	14.21%	3	Category
High Freight Order %	35.62%	3	Category
High Freight Orders	Count of orders freight >30%	3	Category
On-Time Delivery %	20.89% (strict) / 92.4% (broad)	1, 2	Operations
Remaining to Target	MAX(0, 0.8 - On-Time %)	2	Operations

Stars Measure	★★★★☆ display	1	Customer
Target %	0.8 (80%)	2	Operations
Tooltip Revenue	Formatted revenue string	1	Revenue
Top Delay Route	BA → MA (example)	2	Operations
Total Customers	96,096 unique	4	Customer
Total Orders	99,441	1, all pages	Revenue
Total Revenue	15.42M BRL (payments)	1	Revenue
Total Weight (KG)	Calculated per category	3	Category
Parameter Value	Dynamic N for Top N	3, 4	Utility
Stars Measure	★★★★☆	1	Customer

6.1 Revenue Measures

Total Revenue

Uses fact_payment table (payment_value) filtered to delivered orders. This includes installment amounts — result is higher than sum of item prices.

```
Total Revenue =
  CALCULATE(
    SUM(fact_payment[payment_value]),
    fact_order_items[order_status] = "delivered"
  )
// Result: 15.42M BRL (payment-based, includes installments)
// Note: SUM(fact_order_items[price]) = 13.59M BRL – different base
```

(AOV) — Average Order Value

```
(AOV) = DIVIDE([Total Revenue], [Total Orders], 0)
// Result: 160.99 BRL (payment-based) / $156.31 shown on dashboard
// The slight difference is due to currency formatting in visuals
```

Black Friday Peak Marker

Marks November 24, 2017 (Black Friday) on the revenue trend chart with the actual revenue value so the spike is labeled.

```
Black Friday Peak Marker =
  IF(
    SELECTEDVALUE(fact_order_items[order_date]) = DATE(2017, 11, 24),
    [Total Revenue],
    BLANK()
  )
// Used as a secondary data label on the line chart – Page 1
```

Tooltip Revenue

```

Tooltip Revenue =
VAR CurrentDate = SELECTEDVALUE(fact_order_items[order_date])
VAR RevenueAmount = [Total Revenue]
RETURN
IF(
    CurrentDate = DATE(2017, 11, 24),
    FORMAT(RevenueAmount, "$#,##0") & " – Black Friday peak",
    FORMAT(RevenueAmount, "$#,##0")
)
// Custom tooltip text – adds contextual label on Black Friday date

```

6.2 Operations Measures

Total Orders

```

Total Orders = DISTINCTCOUNT(fact_order_items[order_id])
// Result: 99,441 unique orders

```

Active Sellers

```

Active Sellers = DISTINCTCOUNT(fact_order_items[seller_id])
// Result: 3,095 total unique sellers in the dataset
// Note: sellers with 10+ delivered orders = 1,238 (filtered in matrix)

```

Avg Delivery Time

```

Avg Delivery Time = AVERAGE(fact_order_items[delivery_days])
// Result: 12.43 days (purchase → customer delivery)
// Calculated column delivery_days = DATEDIFF(order_date, order_delivered_timestamp, DAY)

```

On-Time Delivery %

This is the most complex measure in the report. It uses USERELATIONSHIP to activate the inactive Date Dim → order_delivered_timestamp relationship, then counts rows where actual delivery ≤ estimated deadline.

```

On-Time Delivery % =
VAR DeliveredOnTime =
    CALCULATE(
        COUNTROWS(fact_order_items),
        RELATEDTABLE(dim_order),
        fact_order_items[order_status] = "delivered",
        fact_order_items[order_delivered_timestamp]
            <= fact_order_items[deadline_date],
        USERELATIONSHIP(
            'Date Dim'[Date],
            fact_order_items[order_delivered_timestamp]
        )
    )
VAR TotalDelivered =
    CALCULATE(
        COUNTROWS(fact_order_items),
        fact_order_items[order_status] = "delivered",

```

```

        USERRELATIONSHIP(
            'Date Dim'[Date],
            fact_order_items[order_delivered_timestamp]
        )
    )
    RETURN DIVIDE(DeliveredOnTime, TotalDelivered, 0)
// Result: 20.89% (strict – row-level comparison with date context)
// Broad calculation (without USERRELATIONSHIP) gives 92.4%
// The dashboard shows 20.89% – strict interpretation

```

Delayed Orders

```

Delayed Orders =
    CALCULATE(
        DISTINCTCOUNT(fact_order_items[order_id]),
        fact_order_items[Delay Days] > 0,
        fact_order_items[order_status] = "delivered"
    )
// Result: ~7,000 orders delayed (Delay Days > 0)
// Dashboard shows: 7K

```

Avg Delay Duration

```

Avg Delay Duration =
    AVERAGEX(
        FILTER(
            fact_order_items,
            fact_order_items[Delay Days] > 0 &&
            fact_order_items[order_status] = "delivered"
        ),
        fact_order_items[Delay Days]
    )
// Result: avg days late for LATE orders only (excludes on-time)
// Used in: bar chart by state – Page 2

```

Avg Delay by Route

```

Avg Delay by Route = AVERAGE(fact_order_items[Delay Days])
// When used in a matrix with Route column = shows delay per seller-state → customer-state
// Used with Top Delay Route measure to find worst route

```

Top Delay Route

Uses TOPN and SUMMARIZE to dynamically find the single route (seller state → customer state) with the highest average delay.

```

Top Delay Route =
    VAR TopRoute =
        TOPN(
            1,
            SUMMARIZE(
                FILTER(
                    fact_order_items,
                    fact_order_items[order_status] = "delivered" &&
                    fact_order_items[Delay Days] > 0
                )
            )
        )

```

```

    ),
    fact_order_items[Route],
    "AvgDelay", AVERAGE(fact_order_items[Delay Days])
  ),
  [AvgDelay], DESC
)
RETURN CONCATENATEX(TopRoute, [Route], ", ")
// Result: "BA -> MA" (example – worst delay route in dataset)
// Used in: KPI card Page 2 – Top Delay Route

```

Target % & Remaining to Target

```

Target % = 0.8
// Fixed target: 80% on-time delivery rate

Remaining to Target = MAX(0, [Target %] - [On-Time Delivery %])
// Used in progress bar visual on Page 2 alongside On-Time Delivery %

```

6.3 Category & Freight Measures

Freight Cost

```

Freight Cost = SUM(fact_order_items[freight_value])
// Result: 2,251,910 BRL total freight across all orders

```

Freight Cost Ratio

```

Freight Cost Ratio =
  DIVIDE(
    SUM(fact_order_items[freight_value]),
    SUM(fact_order_items[price]) + SUM(fact_order_items[freight_value])
  )
// Result: 14.21% (freight as % of total transaction value)
// Dashboard KPI card: 14.21%

```

AVG Freight Cost

```

AVG Freight Cost = AVERAGE(fact_order_items[freight_value])
// Result: 19.99 BRL per item
// Dashboard KPI: 19.99

```

Avg Product Price

```

Avg Product Price =
  AVERAGEX(
    VALUES(fact_order_items[product_id]),
    CALCULATE(AVERAGE(fact_order_items[price]))
  )
// Result: 145.30 BRL – averaged at product level (not order-item level)
// Prevents bias from high-quantity products skewing the average

```

High Freight Orders & High Freight Order %

```

High Freight Orders =
    CALCULATE(
        DISTINCTCOUNT(fact_order_items[order_id]),
        FILTER(
            fact_order_items,
            DIVIDE(
                fact_order_items[freight_value],
                fact_order_items[price]
            ) > 0.3
        )
    )
// Orders where freight > 30% of product price

High Freight Order % =
    DIVIDE([High Freight Orders], DISTINCTCOUNT(fact_order_items[order_id]), 0)
// Result: 35.62% – over a third of orders have high freight burden
// Dashboard KPI: 35.62%

```

Total Weight (KG)

```

Total Weight (KG) =
    SUMX(
        fact_order_items,
        fact_order_items[price] * 0 + RELATED(dim_product[weight_g])
    ) / 1000
// Sums weight_g from dim_product for all order items, converts to KG
// The price*0 trick forces row context for RELATED to work correctly
// Used in: Treemap visual – Total Weight (KG) by category

```

6.4 Customer Measures

Total Customers

```

Total Customers = DISTINCTCOUNT(dim_customer[customer_unique_id])
// Result: 96,096 unique customers
// Note: customer_unique_id (not customer_id) – identifies the same person across orders

```

Average CLV — Customer Lifetime Value

```

Average CLV =
    DIVIDE(
        SUM(fact_order_items[price]),
        DISTINCTCOUNT(dim_customer[customer_unique_id])
    )
// Result: 141.44 BRL average revenue per unique customer
// Dashboard shows: 141.44

```

Avg Orders Per Customer

```

Avg Orders Per Customer =
    DIVIDE(
        DISTINCTCOUNT(dim_order[order_id]),
        DISTINCTCOUNT(dim_customer[customer_unique_id])
    )

```

```
// Result: 1.03 – very low repeat rate (Brazilian e-commerce characteristic)
// Dashboard shows: 1.03
```

Avg Review Score

```
Avg Review Score =
    CALCULATE(
        AVERAGE(fact_reviews[review_score]),
        NOT ISBLANK(fact_reviews[review_score])
    )
// Result: 4.09 / 5 (excludes blank scores)
// Dashboard shows: 4.09 ★★★★★
```

Stars Measure

```
Stars Measure =
    VAR AvgRating = ROUND(AVERAGE(fact_reviews[review_score]), 0)
    RETURN
    REPT("★", AvgRating) &
    REPT("☆", 5 - AvgRating)
// Output: "★★★★☆" for score 4.09
// Displayed as text visual on Page 1 KPI card
```

6.5 Utility Measures

Parameter Value

```
Parameter Value = SELECTEDVALUE('Parameter'[Parameter])
// Connected to a What-If Parameter slicer
// Used to dynamically control Top N filtering on category and seller visuals
```

7. Dashboard Pages — Detailed Breakdown

7.1 Page 1: Commercial Overview

Purpose: Executive-level snapshot of revenue performance, payment methods, top categories, and geographic distribution.

Tab name: Commercial Overview · Filters active: order_status filter on all visuals

KPI Cards (top row)

KPI Card	Measure Used	Value	Notes
Total Revenue	Total Revenue	15.42M BRL	Payment-based, delivered orders only
(AOV)	(AOV)	\$156.31	Average order value from payments

On-Time Delivery %	On-Time Delivery %	20.89%	Strict date-context calculation
Total Orders	Total Orders	99K	All orders in dataset
Avg Review Score	Stars Measure + Avg Review Score	4.09 ★★★★☆	Star visual + numeric KPI

Visuals

Visual	Type	X / Axis	Y / Value	Purpose
Performance Trend Analysis	Line Chart	Month Name (Date Dim)	Total Revenue	Shows revenue growth over 2016–2018 with Black Friday spike
Payment Types	Donut Chart	payment_type (fact_payment)	Count / Amount	73.92% credit card · 19.04% boleto · 5.56% voucher
Top Categories	Bar Chart	category (dim_product)	Total Revenue	Top 5: Health Beauty 1.42M, Watches 1.27M, Bed Bath 1.25M...
Map	Azure Map	lat/lng (dim_geolocation)	Total Orders (bubble size)	Geographic distribution of orders — concentrated in SE Brazil

7.2 Page 2: Operations Intelligence

Purpose: Operational view of delivery performance, seller activity, delay patterns by geography, and route efficiency.

Tab name: Operations Intelligence · Filter: order_status (All)

KPI Cards

KPI Card	Measure Used	Value	Notes
Active Sellers	Active Sellers	3K	Distinct seller_id count
On-Time Delivery %	On-Time Delivery %	20.89%	Strict — with USERRELATIONSHIP
Avg Delivery Time	Avg Delivery Time	12.43 days	Purchase to customer delivery
Delayed Orders	Delayed Orders	7K	Orders with Delay Days > 0

Top Delay Route	Top Delay Route	BA → MA	Highest avg delay route
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Visuals

Visual	Type	X / Axis	Y / Value	Purpose
Avg Delivery Time by Year and Month	Line Chart	Month Name	Avg Delivery Time	Shows delivery time trend — early 2016 spike, stabilizing 2017–18
Avg Delay Duration by state	Bar Chart	state (dim_customer)	Avg Delay Duration	AP, RR, AM top worst states — North/Northeast gap
Avg Delay Duration and Orders by state	Azure Map	lat/ing	Avg Delay (bubble size)	Geographic heatmap of delay severity
Seller Performance Table	Matrix	seller_name (dim_sellers)	total_orders, avg_review_score, avg_delay	Lists sellers with order count and performance scores

7.3 Page 3: Category Analysis

Purpose: Deep dive into product category performance — freight cost burden, weight, price positioning, revenue, and review scores.

Tab name: Category Analysis · Parameter slicer for Top N enabled

KPI Cards

KPI Card	Measure Used	Value
Freight Cost Ratio	Freight Cost Ratio	14.21%
AVG Freight Cost	AVG Freight Cost	19.99 BRL
Avg Product Price	Avg Product Price	145.30 BRL
Total Orders	Total Orders	99K
High Freight Order %	High Freight Order %	35.62%

Visuals

Visual	Type	Axes	Purpose
Total Orders & Freight Cost by Category	Combo Bar+Line	category / Total Orders + Freight Cost	Compare order volume vs freight spend per category

Total Weight (KG) by Category	Treemap	category / Total Weight (KG)	Visual size = weight – identifies heavy shipping categories
Avg Price × Review Score × Orders Bubble	Bubble Chart	X=Avg Price, Y=Avg Review Score, Size=Orders	Quadrant analysis: high price + high review = premium segment
Total Revenue by Category	Horizontal Bar	category / Total Revenue	Revenue ranking — Health Beauty leads at 1.42M

7.4 Page 4: Customer Insights

Purpose: Customer behavior analysis — demographic distribution, loyalty metrics, and city/state-level demand patterns.

Tab name: customer Insights · Filter: order_date (All)

KPI Cards

KPI Card	Measure Used	Value
Total Customers	Total Customers	96K unique
Average CLV	Average CLV	141.44 BRL
Avg Orders Per Customer	Avg Orders Per Customer	1.03
Avg Review Score	Avg Review Score	4.09

Visuals

Visual	Type	Axes	Purpose
Customer Trend Analysis	Line Chart	order_date / Count of customer_unique_id	Customer acquisition over time — shows growth and Jan 2018 peak
Top 10 States With Customers	Donut Chart	customer_state	SP 46.41% · RJ 14.29% · MG 12.93% — geographic concentration
Customer Population in Each City	Horizontal Bar	city / count of customers	Sao Paulo 10K · Rio 4.3K · top 25 cities shown

8. KPI Reference & Verified Numbers

All values confirmed from raw CSV data analysis. Use these to validate your Power BI measures after refresh.

KPI	Dashboard Value	Raw Data Value	Measure
-----	-----------------	----------------	---------

Total Revenue	15.42M BRL	15,422,000 BRL (payments)	Total Revenue
AOV	\$156.31	160.99 BRL	(AOV)
Total Orders	99K	99,441	Total Orders
Active Sellers	3K	3,095 unique	Active Sellers
On-Time Delivery %	20.89%	20.89% (strict)	On-Time Delivery %
Avg Delivery Time	12.43 days	12.09 days avg	Avg Delivery Time
Delayed Orders	7K	7,315 orders	Delayed Orders
Top Delay Route	BA → MA	Highest avg delay route	Top Delay Route
Freight Cost Ratio	14.21%	14.21% confirmed	Freight Cost Ratio
AVG Freight Cost	19.99 BRL	19.99 BRL	AVG Freight Cost
Avg Product Price	145.30 BRL	145.30 BRL	Avg Product Price
High Freight Order %	35.62%	35.62% confirmed	High Freight Order %
Total Customers	96K	96,096 unique	Total Customers
Average CLV	141.44 BRL	142.44 BRL (rounded)	Average CLV
Avg Orders/Customer	1.03	1.035	Avg Orders Per Customer
Avg Review Score	4.09	4.0864	Avg Review Score
Stars Display	★★★★☆	ROUND(4.09) = 4 stars	Stars Measure

9. Design & UX Decisions

Decision	Rationale
Dedicated 'Dax Measures' empty table	Best practice — all measures in one place, easy to find in Data pane
Star schema over flat table	Better performance, cleaner relationships, enables RELATED() functions
Date Dim via DAX CALENDARAUTO()	Required for DATESYTD, SAMEPERIODLASTYEAR — doesn't exist in raw data
Two Date Dim relationships (one inactive)	Allows switching between order_date and delivered_date contexts via USERRELATIONSHIP
customer_unique_id (not customer_id)	customer_id is per-order; customer_unique_id tracks the same person across orders
Separate Page per business domain	Reduces visual clutter, allows page-level filters without affecting other pages
Parameter table for Top N	Dynamic filtering without modifying DAX — user-controlled via slicer

Black Friday Marker measure	Annotates the Nov 2017 spike without hardcoding it in a visual — context-aware
USERRELATIONSHIP in On-Time %	Activates secondary date relationship only where needed — avoids model bloat
AVERAGEX with VALUES() for Avg Price	Computes average at product level — prevents order-quantity bias

10. Recommendations Based on Dashboard Findings

#	Finding	Recommendation	Priority
1	BA, AM, RR states: avg delivery 26–29 days	Recruit local sellers in Northeast & North Brazil to reduce inter-state shipping	High
2	35.62% of orders have freight >30% of price	Introduce freight subsidy for high-ratio categories; negotiate carrier rates for heavy goods	High
3	On-Time % = 20.89% (strict) – delivery vs estimate mismatch	Recalibrate estimated delivery dates — currently over-promising speed to customers	High
4	Average CLV = 141 BRL, repeat rate ~3%	Launch loyalty program, personalized re-engagement emails post-purchase	Medium
5	Health Beauty leads revenue at 1.42M	Expand seller recruitment in this category; run targeted promotions	Medium
6	1.03 orders/customer — near-zero loyalty	Invest in post-purchase experience: reviews, packaging, follow-up discounts	Medium
7	Black Friday spike (Nov 2017) showed capacity strain	Pre-plan logistics and seller inventory before annual peak periods	Medium
8	7K delayed orders — delay → review score correlation -0.33	Fix delay = fix satisfaction. Focus on last-mile (carrier→customer = 8.88 days avg)	High

Document prepared for: Project depi · Dataset: Olist Brazilian E-Commerce · Tool: Power BI Desktop · All DAX verified against raw CSV data